Kagemura: Ninja Clans

Everything that still needs to be built or fixed — a checklist

Reviewed 2026-11-13 at commit **2174d05** (main). Sources: the live code, every audit in docs/audit/ and docs/store/WORLDWIDE_AUDIT.md, the balance audit, the fresh-slate review, the PvP roadmap, SHIPPING/RELEASE/TESTING checklists, the legal and localization docs, and a fresh test and build run. Items already closed in code are left out; items whose closure left a stated limitation are kept.

Build health today

npm test	PASS — 160 files, 1,371 tests passed, 14 skipped (Postgres/Redis tests need TEST_DATABASE_URL / TEST_REDIS_URL).
npm run build	PASS — game chunk 1,147 kB (359 kB gzip) + Phaser chunk 661 kB (175 kB gzip).
Server typecheck + build	PASS.
Content live	Warriors 50/50 · Heroes 1/4 · Buildings 25 of 54 · Keep levels 15 · Store products 14.
Languages	19 of 50 ready (en es pt fr de zh-Hans zh-Hant ja ko hi vi th it nl ro sv ca da nb); 31 at 0%.
Art pipeline	0 of 608 icons in docs/art/ICON_MANIFEST.json produced; everything is drawn in code.
Never done anywhere	A physical phone test, a signed store build, a deployed server, a store review, a legal review, a real player.

Who does it (tag on every item)

CODE An engineering change in this repository. (186 items)
DESIGN A game-design or balance decision first, then usually code. (61 items)
ART Needs a human artist, composer or writer. (18 items)
OWNER Accounts, money, consoles, hosting, business decisions: only the owner can do it. (98 items)
LEGAL Needs a lawyer or a legal decision. (24 items)
DEVICE Must be checked on a physical phone or a signed store build. (22 items)
DOCS Documentation that is wrong or stale. (6 items)

Sections

1. Owner identity, accounts and legal setup (blocks every store listing) — 28 items
2. Store accounts, products and submission — 24 items
3. Server hosting and operations (blocks online play) — 15 items
4. Bugs, exploits and server hardening (code) — 50 items
5. Cheap-phone performance and runtime — 23 items
6. Physical device and signed-build testing — 17 items
7. Store-API and compliance code — 8 items
8. Localization (31 languages to go) — 20 items
9. Native app features to build — 6 items
10. Game content to build — 17 items
11. Balance and game design — 25 items
12. Economy and monetization — 13 items
13. Art, GUI and animation — 20 items
14. Code quality and tech debt — 19 items
15. Growth, retention and live-ops features — 10 items
16. Marketing and launch plan — 8 items
17. Release gates (from RELEASE.md and TESTING.md — none checked yet) — 9 items
18. Mainland China release (separate track) — 27 items

339 items in total. Priority runs top to bottom: sections 1–3 block any launch, 4–9 block a good launch, 10–14 make the game competitive and keep it maintainable, 15–17 are the owner and launch work, 18 is the separate mainland China track.

1. Owner identity, accounts and legal setup (blocks every store listing)

Nothing in code can substitute for these. Most other release gates wait on them.

Identity and name

- [ ] Decide the publisher (individual vs company/LLC), open a bank account, prepare tax details for both stores. — **OWNER**
- [ ] Trademark and store-name search for "Kagemura" (USPTO, EUIPO, JPO; Nice classes 9 and 41); pick a fallback name. — **OWNER LEGAL**
- [ ] Choose the final bundle id / Android package (a Play package can never change). Today it is still com.toadstoolkingdoms.app in capacitor.config.ts:11, android/app/build.gradle, strings.xml and the Xcode project. — **OWNER**
- [ ] After the id is chosen: update Capacitor config, Gradle namespace/applicationId, Java package folder, Xcode PRODUCT_BUNDLE_IDENTIFIER, server APPLE_BUNDLE_IDS / GOOGLE_PLAY_PACKAGE, IOS_BUNDLE_ID secret. — **CODE**
- [ ] Register a domain; decide the hosting region and the crash-report (Sentry) region. — **OWNER**
- [ ] Turn off or redirect the old Toadstool Kingdoms GitHub Pages site: it shares the same origin, IndexedDB, localStorage and caches with Kagemura (each service worker deletes the other's caches). — **OWNER**

Publisher facts (public/legal-config.js — every field is empty today)

- [ ] Fill operator, operatorAddress, operatorPhone, responsiblePerson, supportEmail, deletionUrl. — **OWNER**
- [ ] Fill euRepresentative, ukRepresentative, hosting, crashReportRegion, governingLaw. — **OWNER LEGAL**
- [ ] Fill the four retention periods (support mail, deletion requests, inactive accounts, server logs) and termsEffective. — **OWNER LEGAL**
- [ ] Remove the test that asserts these fields are empty once they are filled; replace "[deletion URL]" in docs/store/REVIEW_NOTES.md. — **CODE**
- [ ] Make the server deletion page title match the Play listing name ("Kagemura: Ninja Clans"). — **CODE**

Legal decisions and reviews

- [ ] Lawyer review of privacy policy, terms and community rules per market (GDPR Art. 13, CCPA notice at collection, LGPD encarregado, automated-decision note, processor regions, UK Children's Code DPIA). — **LEGAL**
- [ ] Decide the under-13 strategy (offline-only, full COPPA/Families programme, or internal-operations-only data). Today under-13 players get online accounts, a persistent Player ID, possibly push tokens and crash reports, and can read adults' chat, while Play is declared 13+. — **LEGAL OWNER**
- [ ] Then implement it in code (gate push tokens, crash reports and adult chat by age band). — **CODE**
- [ ] Germany: Impressum; German-law review of the terms. — **LEGAL OWNER**
- [ ] Japan: fill seller facts on /ja/tokushoho.html and /ja/shikin-kessai.html; review terms §3; base-date calendar and deposit plan. — **LEGAL OWNER**
- [ ] UK Online Safety Act: name a responsible person, risk assessment, response targets, complaints procedure, crisis escalation note. — **LEGAL OWNER**
- [ ] Brazil: ECA Digital (self-declared age not accepted for minors' social features), legal representative, LGPD DPO. — **LEGAL OWNER**
- [ ] India: IT Rules grievance officer with contact and response commitment; DPDP review. — **LEGAL OWNER**
- [ ] EU: decide whether to show real-money equivalents next to Jade prices (CPC virtual-currency principles); declare DSA trader status in both consoles. — **LEGAL OWNER**
- [ ] European Accessibility Act position: micro-enterprise exemption, or an accessibility statement and WCAG 2.1 AA purchase flow. — **LEGAL**
- [ ] Decide the Apple CONSUMPTION_REQUEST policy (answer or not). — **OWNER LEGAL**
- [ ] Define a "significant update" for the iOS age-range acknowledgement flow. — **OWNER LEGAL**
- [ ] Legal check of the Season Scroll "≈ N Jade value" claim and the intent-less-receipt hold wording. — **LEGAL**

☐	Korea GRAC rating / domestic-agent thresholds; Texas SKU age ratings.	OWNER LEGAL
☐	Mainland China is a separate release track: see section 18. Until its licence exists, untick China mainland in App Store Connect → Pricing and Availability (Google Play does not serve mainland China).	OWNER
☐	Other territories: exclude Vietnam (no licensed partner), Indonesia (until PSE registration), Taiwan if unlicensed, sanctioned markets; expect no payments in Russia/Belarus.	OWNER LEGAL
☐	Decide whether to keep the name "Jade"; native Japanese cultural review of names, kamon, buildings and the mentor.	OWNER

2. Store accounts, products and submission

Apple and Google console work, then the first signed builds.

Apple

☐	Apple Developer Program: Paid Apps agreement, bank, tax, Small Business Program.	OWNER
☐	New App ID with Sign in with Apple, Game Center, Push and Declared Age Range; keys for SIWA, App Store Server API, APNs.	OWNER
☐	Register server-notification URLs (production and sandbox) and the SIWA server-to-server URL.	OWNER
☐	Age-rating questionnaire (Contests: Frequent; Age assurance: Present; Messaging/UGC: Yes); privacy label; export compliance.	OWNER
☐	GitHub secrets for TestFlight: APPLE_TEAM_ID, ASC_KEY_ID, ASC_ISSUER_ID, ASC_KEY_P8_BASE64.	OWNER
☐	Move iOS CI from macos-15 to macos-26 and fail (not warn) below the 26.5 SDK (.github/workflows/ios-testflight.yml).	CODE
☐	Run the MARKETING_VERSION workflow on a macOS runner once.	OWNER

Google Play

☐	Play Console identity verification, payments profile, the new package.	OWNER
☐	Upload keystore + Play App Signing; set ANDROID_KEYSTORE_BASE64 / _PASSWORD / KEY_ALIAS / KEY_PASSWORD so the aab job actually builds (it is skipped today).	OWNER
☐	Set build vars VITE_SERVER_URL, VITE_PUBLIC_URL, VITE_GOOGLE_CLIENT_ID (+ store URLs, support email, privacy/support/community URLs).	OWNER
☐	Google Cloud OAuth clients (web + Android with Play-signing, upload and debug SHA-1s), published consent screen.	OWNER
☐	Firebase project + google-services.json (no Android FCM push until it exists).	OWNER
☐	Data safety form, account-deletion URL, target-audience decision, IARC questionnaire, UGC declaration.	OWNER
☐	Request Play Age Signals API access; service account for the Play Developer API; RTDN Pub/Sub subscription; license testers.	OWNER
☐	Closed testing with 12+ testers for 14 days (required for new personal accounts).	OWNER

Products and listings

☐	Create all 14 in-app products in both consoles under the ids in store.json (stipend described as "30 Jade payments", never "monthly").	OWNER
☐	Regional price templates and a low-income-market pricing policy (one USD ladder today).	OWNER DESIGN
☐	Generate store screenshots (npm run store:screenshots) and crop the sandbox note off the Store shot; caption them.	CODE OWNER
☐	Make the 1024×500 Play feature graphic and a 15–30 s vertical preview video.	ART OWNER
☐	Rewrite keywords (duplicates, off-theme); recheck listing text against the commercial build (GLOBAL-020).	OWNER
☐	Seed an Alliance with members and recent chat, and some low-Keep players, for App Review.	OWNER
☐	Write store listings for the 31 unready languages as they ship (docs/store// only exists for 18).	OWNER

- [ ] Submit with phased rollout; plan 48-hour review responses; watch "all bots" and "pay to win" themes. — **OWNER**
- [ ] Add VITE_APP_STORE_URL / VITE_PLAY_STORE_URL to the RELEASE.md checklist. — **DOCS**

3. Server hosting and operations (blocks online play)

No server has ever been deployed. The CI deploy and backup jobs exit 0 with a warning when their secrets are missing.

- [ ] Choose and pay for Postgres with point-in-time recovery, Redis, two API instances, object storage for backups, Sentry, monitoring. — **OWNER**
- [ ] Set every production secret: DATABASE_URL, REDIS_URL, DEPLOY_ENV, ADMIN_TOKENS, ADMIN_ALLOW_IPS, METRICS_TOKEN, SENTRY_DSN, SUPPORT_CONTACT, CORS_ORIGIN, TRUST_PROXY, MIN_CLIENT_BUILD, Apple/Google/APNs/FCM keys, GOOGLE_RTDN_*. — **OWNER**
- [ ] Enable Sentry "Prevent Storing of IP Addresses" (GLOBAL-019). — **OWNER**
- [ ] Point api. with A and AAAA records; verify HSTS and IPv6 with tools/ops/check-domain.mjs. — **OWNER**
- [ ] Create the staging and production GitHub environments with DEPLOY_HOOK, SERVER_URL and required reviewers so deploys and contract tests actually run. — **OWNER**
- [ ] Turn on GitHub Pages (Source: GitHub Actions). — **OWNER**
- [ ] Move backups off GitHub artifacts to an owned bucket (PITR/WAL); set BACKUP_DATABASE_URL and BACKUP_KEY; run one real restore into a fresh database. — **OWNER**
- [ ] Configure the host load balancer to poll /readyz with a kill timeout above the shutdown deadline (A9). — **OWNER**
- [ ] Run server:bench and server:load on the real host; size workers, queue, instances, rate limits; record the numbers. — **OWNER**
- [ ] Try outbound timeouts against real Google Play, FCM and JWKS endpoints (A10). — **OWNER**
- [ ] Pager for the alert rules, on-call rota, status page, weekly review of desyncs, refunds, forfeits and staff actions. — **OWNER**
- [ ] Staff at least one trained moderator and a support mailbox; written refund, chargeback and cheat-ban policies. — **OWNER**
- [ ] Deploy the server from the same commit as the first store build, migrations first (including 0029 on external Postgres). — **OWNER**
- [ ] Build the slimmed Docker image and confirm it; add HEALTHCHECK and an init process; pin the base image (A26). — **CODE**
- [ ] BACKEND confirmation of the powerSlackAt matchmaking change (docs/backend/REQUESTS.md, 2026-11-06). — **CODE**

4. Bugs, exploits and server hardening (code)

Confirmed or open defects found in code. Items marked "partly" were reduced by earlier fixes but not closed.

Player-visible bugs

- [ ] "What is shared" statistics dialog shows English in every language: src/main.ts:329 passes STATS_EXPLAINER raw to hud.confirm() (hud.ts:2046 wraps it correctly with i18nLabel). — **CODE**
- [ ] Scout chip text "AA" / "splash" is keyed only in the aria label (F16). — **CODE**
- [ ] Battle HUD drops scout tips that name warriors the player did not bring: scoutReport needs the attacker's army (battleHud.ts:361 TODO; game-theory REQUESTS). — **CODE**
- [ ] Train top-up then cancel refunds 100% into storage, converting Jade into Provisions (CU-3, cancelTraining). — **CODE**
- [ ] Season Scroll and calendar rewards scale with the Keep at claim time: waiting to claim pays up to ×1.9 (CU-6 / E07). — **CODE**
- [ ] Refunds and rewards are lost at full storage; dismiss refund uses the current level's price (CU-7, P8). — **CODE**
- [ ] Weekly "clear 2 Trial tiers" task becomes impossible past tier 60; taskEligible ignores Trials (E27). — **CODE**
- [ ] Collector upgrades stop production and the quote does not say so (P5). — **CODE**

- [ ] The "20% raidable" rule and the loot cap are shown nowhere. **CODE**
- [ ] Abandoning an offline battle keeps the scout: a free probe (E16, re-check). **CODE**
- [ ] Daily chest is predictable from a choosable seed on device villages (E18). **CODE**
- [ ] Refund debt shows as a plain negative number in the HUD (A5). **CODE**
- [ ] Refused cross-device actions are not listed in the HUD (GLOBAL-009). **CODE**
- [ ] Age question: "Not now" is never asked again (E02). **CODE DESIGN**
- [ ] Stipends and Scrolls bought before save v20 are refunded with approximations. **CODE**

Economy integrity and anti-cheat

- [ ] Loss hiding (A1, partly): the seed predicts win/loss ~98% of the time, so offline cherry-picking is still possible (now costs Next). **CODE DESIGN**
- [ ] Device-village progress is client-trusted until import (A2/E11/E21, partly): accounts registered early and left to age still pass. Decide whether to create online villages server-side at first contact. **OWNER DESIGN CODE**
- [ ] flags.sandbox can be removed by editing a save code; holds have no staff release button (GLOBAL-002). **CODE**
- [ ] Refund pause counted per player id, not per store identity (A5). **CODE**
- [ ] Device-village refund settlement can be ignored by a modified game (needs server online). **CODE DESIGN**
- [ ] A modified client can release a purchase hold and then pay (GLOBAL-005). **CODE**
- [ ] Under-18 cap and analytics count catalog USD, not local store proceeds (GLOBAL-007). **CODE**
- [ ] Online Alliance War chests still pay members who did not attack (E14); bot Gang War resource chests unchanged (E12). **CODE DESIGN**
- [ ] Free rerolls on device villages pay Next only on the device (E15); run migration 0025 on Postgres. **CODE**

Server security and robustness

- [ ] Trial / Bounty / rematch battles simulate on the request thread inside a row-locked transaction; move them to the worker pool (A3). **CODE**
- [ ] Per-player rate limits keyed by token hash while players can mint unlimited tokens (A7, app.ts:215). **CODE**
- [ ] Limiter fallback is per instance: N instances allow N× the limit (A8). **CODE**
- [ ] Google purchase check trusts the client product id (A11, re-check). **CODE**
- [ ] Idempotency scope is token + key only, ignoring method/route/body (A13). **CODE**
- [ ] Sign-in tokens not nonce-bound; Game Center cert chain and dates not checked (A15). **CODE**
- [ ] Bearer tokens never expire or rotate; no "sign out this device" (A16). **CODE**
- [ ] Duel partners not counted as a tie for friend requests (A14); village routes keep idempotent answers for a day (A12). **CODE**
- [ ] Apple notifications: no UUID storage or ordering; account-delete notification deletes at once (A29). **CODE**
- [ ] Names and friend codes (PII) appear in URL logs (A30). **CODE**
- [ ] TRUST_PROXY hop misconfiguration risk; add an IP-echo endpoint (A31). **CODE OWNER**
- [ ] Re-sim timeouts not counted against the ticket; client x-request-id trusted (A32/A33). **CODE**
- [ ] No staff-auth-failure metric or alert (A28). **CODE**
- [ ] A periodic job that outlives its lease can overlap a takeover (A21). **CODE**
- [ ] Anonymous accounts never pruned; no dormancy policy (A23); sandbox purchase cap is per free account (A24). **CODE LEGAL**
- [ ] Contact/link filter misses other apps' spellings, other languages and handles spelled in words (C25); links in chat and names are not stripped (UK OSA). **CODE**
- [ ] Staff cannot rename an offending player name; no compliance-complaint route in support.html. **CODE**
- [ ] Pin CI actions by SHA; add npm audit / SBOM step (A27). **CODE**

Server scale (fix before 100k MAU)

- [ ] Whole-village JSONB rewrite (~290 KB) on every command batch (A17). CODE
- [ ] Retention deletes unbatched under the 30 s statement timeout; will silently stop (A18). CODE
- [ ] Matchmaking uses ORDER BY random() over a widening window (A19, pgRepo.ts:924). CODE
- [ ] Power leaderboard is a full scan per call (A20). CODE
- [ ] Migrations cannot build indexes concurrently (no no-transaction header) (A22). CODE
- [ ] Polling cost for Alliances: measure payloads, add deltas/compression (R04). CODE DEVICE
- [ ] Real load test (current one is a stub); test A7, A11, A13, A15, A18, A19 and import plausibility at scale (A34); cap concurrency across processes (GLOBAL-004). CODE
- [ ] Single region: a sync is ~6 sequential calls (2–3 s from Jakarta/Lagos) — batch calls or add regions (I01). CODE OWNER

5. Cheap-phone performance and runtime

Every runtime fix so far was verified only in Chromium emulation.

- [ ] Texture budget covers building art only; add the island, scenery, warriors and badges (F1 / M1). CODE
- [ ] Island painter is one ~4 s main-thread task at CPU×6; split it. CODE
- [ ] lazyTiers flag is dead (_lazyTiers unused in assets.ts); the advertised low-profile saving does not exist (F16). CODE
- [ ] Quality thresholds are guesses; the downgrade applies only at next start; 120 Hz phones judged by the same 40 FPS line; HUD FPS readout over-reports (F2, R03). CODE DEVICE
- [ ] Boot "Retry" still reloads mid-paint on slow networks (F3). CODE
- [ ] Network-first navigation waits 3.5 s on every slow-link launch (F4, NAV_TIMEOUT_MS). CODE
- [ ] Service worker re-fetches 512 px icons, Apple icon and legal pages on every build (F5 / M4). CODE
- [ ] Render-blocking CSS gates the splash; no inline boot rules or modulepreload; 1.5 s font wait (F7, F17, M5). CODE
- [ ] fx.ts still uses createCanvas (ImageData copies) for ridges, ruins, fog, villagers, birds (F8). CODE
- [ ] 6–7 full-screen alpha layers per frame; low profile does not cut the count (F10). CODE
- [ ] Synchronous PNG encode of every thumbnail; full-res data URLs (F11, M10). CODE
- [ ] Set android.minWebViewVersion to 80 (the ES2020 bundle needs it); boot message says "Chrome" inside the app (F12). CODE
- [ ] Under the Canvas fallback, tints other than role rings are lost (F13). CODE
- [ ] Autosave serializes the whole village every 15 s with no dirty check (F14, main.ts:647). CODE
- [ ] Damaged-save backups accumulate without limit (F15). CODE
- [ ] No onTrimMemory / iOS memory-warning handling (F19). CODE
- [ ] Village-life object churn on low / reduced-motion (F20). CODE
- [ ] AudioContext created on first tap even with sound off; no AVAudioSession configuration in iOS Swift (F9/F18). CODE
- [ ] Backdrop blur behind modals on "high" while the village animates (M6); HP bars retessellated every frame (M7). CODE
- [ ] Loot chips rebuilt every 100 ms; 3 s toasts too short for long errors (M10). CODE
- [ ] Textures not DPR-aware; island not tiled; no texture atlas; high batch count (Art F5, F20). CODE
- [ ] Code-split Store, Settings and Codex panels; drop sandbox tools from the production bundle (Q31, M2). CODE
- [ ] Last-write-before-kill can still lose unsent actions (GLOBAL-010). CODE

6. Physical device and signed-build testing

Buy or borrow: a 2 GB Android Go phone, a 3–4 GB Android, a 2019-class iPhone on iOS 15–16, and a current flagship of each.

Performance

- [ ] Measure cold start, peak memory over a 30-minute session (scouting several Keeps, a Trial, a replay), sustained battle FPS, battery per hour; calibrate the quality thresholds from these. **DEVICE**
- [ ] Airplane mode, 3G throttling, low storage, suspend/kill, WebGL disabled (Canvas fallback), context loss. **DEVICE**
- [ ] Real first visit, offline reload and packaged native boot with the art manifest (GLOBAL-011); service-worker behaviour with interrupted downloads and quota (GLOBAL-018). **DEVICE**
- [ ] Safe areas and the Build button on real notched phones (GLOBAL-014); iOS "Home Screen" storage advice on an iPhone (F6). **DEVICE**
- [ ] Check the signed AAB in APK Analyzer for 16 KB page size (Firebase Messaging via the push plugin); record installed AAB/IPA size. **DEVICE**

Purchases and accounts

- [ ] Play Billing 9 matrix: sandbox purchase, pending, cancel, repeat consumption, restore, refund and refund reversal; server consume/acknowledge against the real Play Developer API. **DEVICE**
- [ ] iPhone: purchase, Ask to Buy, refund, reinstall, real storefront prices and currencies (GLOBAL-007); app killed after a failed save (GLOBAL-003). **DEVICE**
- [ ] Build TkViewController plugin registration in Xcode; open Products.storekit in Xcode once. **DEVICE**
- [ ] Sign in with Apple, Google and Game Center: reinstall and second-device recovery (GLOBAL-008). **DEVICE**
- [ ] Play Age Signals access prompt on a Play-installed build; iOS Declared Age Range in Sandbox (Xcode 26.5+). **DEVICE**

Everything else

- [ ] Push delivery, quiet hours and the real system permission prompt. **DEVICE**
- [ ] Android Back, App Links / Universal Links, update path from an older store build. **DEVICE**
- [ ] Listen to sound and music on a phone speaker and headphones (nobody has). **DEVICE**
- [ ] Pinch-zoom, large text, screen reader and reduced motion on real devices. **DEVICE**
- [ ] Glyph and shaping checks for every ready language on cheap phones; IME input for names and chat. **DEVICE**
- [ ] Long offline outages: what the player sees when a raid report expires (R01, R02). **DEVICE** **DESIGN**
- [ ] Run the tester script with TestFlight external testers and the Play closed track. **DEVICE** **OWNER**

7. Store-API and compliance code

- [ ] iOS: write requiredRegulatoryFeatures and the significant-update acknowledgement (PermissionKit) once on the 26.5 SDK; handle RESCIND_CONSENT arriving without a transaction (C02). **CODE**
- [ ] Japan: build the unused-paid-balance (¥10M threshold) staff report; capture Google prices; key Japanese notices on storefront, not device language (C17). **CODE**
- [ ] EU: show real-money equivalents for Jade-priced items if the legal decision says so (C16). **CODE**
- [ ] Age bands: document usage limits of age-band metrics (C11). **CODE** **LEGAL**
- [ ] Add CFBundleLocalizations to Info.plist (C12, re-check). **CODE**
- [ ] Per-language profanity, slur and grooming/contact blocklists (today: 30 English words in social.json; server has none) (C23, R06). **CODE** **OWNER**
- [ ] Terms acceptance flow once termsEffective is set; older game builds cannot chat on a new server (C14). **CODE**
- [ ] Write an offline-play promise into listing/help (R01) and player messaging for raid reports older than 72 h (R02). **CODE** **DESIGN**

8. Localization (31 languages to go)

docs/localization/README.md marks steps L01–L14 as TODO; partial work does not count as done.

Translate

- [ ] Wave 1: Arabic (next), Indonesian, Russian, Turkish. — CODE
- [ ] Wave 2: Polish, Ukrainian, Malay, Bangla, Urdu, Swahili, Hausa. — CODE
- [ ] Wave 3: Yoruba, Amharic, Zulu, Igbo, Marathi, Telugu, Tamil, Gujarati, Kannada, Malayalam, Punjabi. — CODE
- [ ] Wave 4: Czech, Greek, Hebrew, Hungarian, Croatian, Finnish, Slovak, Slovenian, Odia. — CODE
- [ ] Per language, activate: generatedNames in languages.json, server/localization.ts + accountPage.ts, ACCOUNT_FAILURE in tools/i18n/pages.mjs, public/legal.js fallback, iOS .lproj, Android values-xx + locales_config.xml, StoreKit fixtures, docs/store//, tests/locales/.test.ts. — CODE
- [ ] Native-speaker linguistic review of all 19 ready languages and each new one. — OWNER

Text layer gaps

- [ ] Server errors and moderation notices: send an error code + parameters instead of English sentences (~245 sites in server/core; backend REQUESTS). — CODE
- [ ] Sim events (fail(code, message), close-call notes) should carry ids + parameters (game-theory REQUESTS). — CODE
- [ ] Generated computer-player names stay in Latin letters because they are stored text. — CODE DESIGN
- [ ] Re-verify NAME_ERROR_TEXT and server rejection reasons go through the text layer (L06); PRESET_CHAT and notices in each language (C24). — CODE
- [ ] L01: reviewed mapping by inventory id; regenerate INVENTORY.json against the current tree. — CODE
- [ ] L14 gates: full journey matrix, RTL and stretched pseudo-locale runs, coverage report per language × surface × platform. — CODE

Scripts, fonts and RTL

- [ ] RTL: only 4 CSS rules mirror today; canvas text and absolute positions not mirrored; add an RTL pseudo-locale test (L04). — CODE
- [ ] Add data-script rules for Arab/Hebr (letter-spacing normal) and font stacks for Cyrl, Grek, Beng, Telu, Taml, Gujr, Knda, Mlym, Guru, Orya, Ethi, Urdu Nastaliq. — CODE
- [ ] Add new Latin-extended languages (pl, cs, sk, sl, hr, hu, tr, ha, yo, ig…) to SYSTEM_FONT_LANGS (fonts.ts:22). — CODE
- [ ] Long-translation overflow tested with Spanish only (M9); test with German, Finnish and others. — CODE
- [ ] Localized typography review for Japanese, Chinese, Korean, Arabic, Thai, Hindi. — OWNER
- [ ] Translate store listings, screenshot captions, community rules and moderation notices for every launch language; localized help. — OWNER

Stale localization docs

- [ ] docs/localization/README.md:24 says 35 unready languages (real: 31); SPANISH_IMPLEMENTATION.json says "other 48". — DOCS
- [ ] docs/localization/SPANISH_SIZE.json is rewritten by every test/build run; make it deterministic or stop writing it from tests. — CODE

9. Native app features to build

- [ ] App Links / Universal Links: add an appUrlOpen handler, assetlinks.json and apple-app-site-association on the domain, Associated Domains entitlement (R09). Shared links open the web game today. — CODE OWNER
- [ ] Play Games sign-in and friends (optional; no SDK in android/). — CODE
- [ ] Game Center / Play Games achievements (G31). — CODE OWNER
- [ ] In-app review prompt, only after a win (G26). — CODE
- [ ] Accessibility: rem-based type (0 rem units today, 61 px font sizes), sub-44 px tap targets, 10–11 px fixed type, screen-reader menus, canvas alternatives (R10, M8, F21). — CODE
- [ ] Web payments (Stripe) — optional, later; live co-op strike over WebSocket (roadmap step 71) — deferred. — CODE OWNER

10. Game content to build

Heroes (1 of 4 live)

- [ ] Iron Matriarch (Guardian, Keep 9, "bulwark" shield wall blocking defence fire in a cone for 6 s): data, sim ability, art, SIM_RULES bump. — CODE ART
- [ ] Sky Admiral (Sky, Keep 11, "squall": pushes flyers forward, shields them from anti-air 5 s). — CODE ART
- [ ] Grand Sensei (Mystic, Keep 13, "inner_calm": heals 40% and ignores the next jam/trap/splash). — CODE ART
- [ ] Hero Hall (hero_shrine) building; hero choice and levelling; hero upgrades; include the hero in army-lab gates (B5, Growth 5). — CODE DESIGN
- [ ] Fix the Shadow Captain's Rally aura, measured as a no-op (RM-3). — CODE DESIGN

Buildings (29 of 54 not live)

- [ ] Designed effects to implement: whirling_blade_tower (spin AoE), rain_barrel_tower (extinguish), listening_jars (detect underground), living_bamboo_wall (regrow), straw_scarecrow (beast decoy). — CODE DESIGN
- [ ] mirror_screen (reflect projectiles), sand_bucket_post (smother splash), discipline_banner (focus aura), spotter_perch (instant retarget), steam_vent (no-projectile AoE). — CODE DESIGN
- [ ] sleep_dart_tower (sleep), rolling_log_chute (line crush), moonlit_gravel (arrival stun), ink_seal_post (no revive), lightning_rod_pagoda (absorb lightning). — CODE DESIGN
- [ ] No effect designed yet: scroll_library, shadow_ink_well, shadow_ink_vault, reinforced_gate, sky_net, temple_bell_tower, clan_forge, wind_chime_wire, crossbow_bastion, flame_trench, thunder_shrine, ninja_watch_post, great_dragon_cannon. — DESIGN CODE
- [ ] No new buildings after Keep 12; Keep 7–15 caps are projected, not authored (W5). — DESIGN

Systems and modes

- [ ] Warrior Mastery: documented but not implemented (B4). — CODE DESIGN
- [ ] Trial-tower and Gang shared-goal events: specified only. — CODE DESIGN
- [ ] Story premise and chapters; hero and rival-leader bios; named rival clans; 12-month event calendar incl. seasonal content. — ART DESIGN
- [ ] Server-seeded bank of real-looking rival bases for cold start (Growth 12, E29). — CODE
- [ ] Deployment takes 100–200 taps: add a squad / "deploy N" gesture or pause (S2). — CODE DESIGN
- [ ] Defence feedback: heat map, held-rate summary, more than 10 records (S4). — CODE
- [ ] Scouting never shows traps; opponents never upgrade mid-raid (D6). — DESIGN

11. Balance and game design

From the 2026-11-08 balance audit and the 2026-11-06 fresh-slate review, after the round-2 fixes and battle rules 7. All evidence so far is model players and bots, never real people.

Progression and difficulty

- [ ] Casual Keep 5→6 wall of ~26–28 days lands on the D30 window; add a milestone ladder or catch-up track (E23, H7, G18). — DESIGN
- [ ] Casual explorer averages 0.14–0.55 stars at Keeps 8–14; tests only guard against it getting worse (E25). — DESIGN
- [ ] Casual Keep 3: 1.00 stars, explorer 0.60 (E24); Keeps 1–2 offer no choice (P6). — DESIGN
- [ ] Keep levels too fast (Keep 15 ~day 113) yet builder-bound from Keep 8 by the 7-day timer cap (P1, P2, DM-3). — DESIGN
- [ ] Research is the late wall (tierCostMult 3.0); tier-3+ research not priced in Coins; Provisions wall at Keeps 13–15 (P4, CU-1/2). — DESIGN
- [ ] Casual raid tax 1.2–5× loot; shield tiers still 12/14/16 h (LD-4, P3). — DESIGN
- [ ] Hand-written per-Keep tables (six of them) with no derivation test (X3). — CODE

Armies and defence

- [ ] Siege and beast families are dead picks; Shinobi/Ram/Crane bottom picks (AS-3, B1). — DESIGN

- [ ] Shadow alone on the Pareto front; DPS taxes on Thousand Needle, Siege Bombardier, Hawk Tamer (RM-1). **DESIGN**
- [ ] Newer tiers often worse than older ones; Codex presents sidegrades as upgrades (RM-2, B2, B3). **DESIGN CODE**
- [ ] Twelve zero or negative kits (e.g. Bell Warden −5.7); add a "kit ≥ +2 points" lab gate (RM-3). **DESIGN CODE**
- [ ] Suggested army is not layout-dependent (suggest.ts has no archetype input) (AS-1/2). **CODE**
- [ ] Defence does not pay: defenseWin share 0.05, held raid grants no shield; add rusher-vs-turtle campaign gate (DM-2, D1). **DESIGN CODE**
- [ ] Mortar best defence per Coin but capped at 2, level 6 frozen at Keeps 7–10 (DM-1). **DESIGN**
- [ ] Walls irrelevant: count frozen at 125 from Keep 6; ten-tile detour; layouts have gaps (D2, DM-4, CR-1). **DESIGN**
- [ ] Too few defences per Keep (16 for 360 housing); no counter matrix (D3, D4). **DESIGN**
- [ ] Entry side is the biggest lever and point intel does not predict it; closed approaches are a layout hash that one decor tile flips (DZ-1, DZ-2, D5). **DESIGN CODE**
- [ ] Own attack ends the shield, a Trial does not; settle-the-score rematch has negative EV (LD-3). **DESIGN**
- [ ] Honor does not reach the matchmaker; league chests reward session count (LD-2). **DESIGN**
- [ ] 180 s clock punishes second waves; abilities small; global battle constants not data-scaled (CR-2, X5). **DESIGN**
- [ ] Night of Raids saving and adaptive tasks lowered by throwing raids (LD-5). **DESIGN**

Measurement

- [ ] Add paired marginal tests, a balance:search gate, rusher/turtle/sandbagger campaign gates, per-family lower outcome gate in army-lab (Change-12, X2). **CODE**
- [ ] Unmeasured leads: human-planned attacks vs maxed bases; gapless wall rings; jam chain under 18 Ghost Engineers; rating.ts layout-reset dodge; dollars to Keep 15 by paying only. **DESIGN**
- [ ] Independent adversarial re-verification of the balance audit (never run). **DESIGN**
- [ ] Real-player playtests. **OWNER**

12. Economy and monetization

- [ ] Free Jade (~130–540/month) competes with paid packs; time is the only purchase (JD-1, E05 not re-measured). **DESIGN**
- [ ] Dollar ladder inverts: packs 45→75 Jade/$, stipend ~120/$, offers 130–184 (JD-3, E03, H2). **DESIGN**
- [ ] Paying to rush the Keep harms the buyer in matchmaking; no soft Keep gate (E06). **DESIGN**
- [ ] Jade price of resources 37× steeper at Keep 15; top-ups dead after Keep 7 (CU-5). **DESIGN**
- [ ] Tutorial points at Jade "Finish now" first, before the age question (JD-2, H4). **DESIGN CODE**
- [ ] Still one offer per session (E01); offers not keyed to the real walls (collectors/research) (E23). **DESIGN**
- [ ] Thin Jade sinks: no shield purchase, no extra research lane (E19). **DESIGN**
- [ ] Thin cosmetics: no warrior/hero skins, seasonal themes, banners, cosmetic pass tier; looks reuse the Keep painter (H6). **ART DESIGN**
- [ ] Pack ladder has no anchor / "most popular" card, no pack art or hero card (E04, F25). **ART DESIGN**
- [ ] League win bonuses not Keep-scaled (E10). **DESIGN**
- [ ] Calendar rewards not measured against the faucet budget (Growth 19). **DESIGN**
- [ ] Two clocks: chest, stipend, seasons and Alliance War at UTC, tasks at local midnight (H3, Growth 9). **CODE DESIGN**
- [ ] Teens see no offers — model it in revenue forecasts (G21). **DESIGN**

13. Art, GUI and animation

Upper "polished indie" today. The biggest gaps need a human artist.

Commission (human)

- [ ] 4–6 character faces (Sensei Hoshi, playable hero, three rival clan leaders) for icon, splash, store, packs and result screens. **ART**
- [ ] Character-led app icon with A/B variants (today it is a building) (F23). **ART**
- [ ] Animation key poses for the 10 roles (walk, attack, hit, fall); today warriors are single-frame (F6). **ART**
- [ ] Building damage states (ruins read as black discs) (F7); Keep / Dojo / Courtyard concept pass; visible per-level changes (F11). **ART**
- [ ] Painted ground and environment (grass, paths, paddies, cliffs, trees). **ART**
- [ ] Two 60–90 s music tracks (village, battle) plus stingers; battle music is a 7.5 s loop (F17). **ART**
- [ ] Produce the 608 icons in docs/art/ICON_MANIFEST.json (0 produced) or retire the manifest. **ART**
- [ ] Real Keep-skin swatches in Store → Looks (placeholder today) (Art F2). **ART**

GUI code

- [ ] Battle-start zoom still 0.14 fit floor: smaller map frame or collapsible opponent card (Art F4). **CODE**
- [ ] First frame: camera and Sensei bubble still cover the Keep (F13). **CODE**
- [ ] Top HUD covers 25–31% of a 320–360 px screen (F14). **CODE**
- [ ] Battle troop cards use the dark field sprite, not the readable warrior badge (F12). **CODE**
- [ ] Dialogs have no exit animation (F10). **CODE**
- [ ] Overlapping edge chevrons and stacked callouts (F8); army count pill overlaps its button (F22). **CODE**
- [ ] Building sheets show no picture or next-level preview (F26). **CODE**
- [ ] Defences never turn toward their target (F27). **CODE** **ART**
- [ ] Boot card uses the 192 px icon upscaled (F24). **CODE**
- [ ] Ridges unchecked on 21:9 / ultra-wide (F3); Provisions icon colour-blind contrast (F9). **CODE**
- [ ] Unlock ceremony for new warriors (G22); "Collect all" (G33). **CODE**
- [ ] Next-objective chip covers no research or army goals (Growth 7); tutorial end does not point at chest/Scroll (G6). **CODE**

14. Code quality and tech debt

Code health is good (strict TS, 0 any). The debt is in big files, missing tooling and docs.

- [ ] Add a linter and formatter (none exists: no eslint/prettier/biome/.editorconfig) (Q6). **CODE**
- [ ] Turn on noUncheckedIndexedAccess (Q7); validate JSON data types instead of asserting them (Q34). **CODE**
- [ ] Split god files: hud.ts 2,430 lines, ninja.ts 2,648, village.ts 2,033, main.ts boot(), protocol.ts 1,202, connection.ts 1,135 (Q17, Q18). **CODE**
- [ ] Split server/repo.ts (2,067 lines, interface + MemoryRepo) and pgRepo.ts (1,622) (Q16). **CODE**
- [ ] server/app.ts: ~108 inline routes with no schemas and `as never` body casts (Q19). **CODE**
- [ ] Break import cycles (battle↔warriors↔fortifications, pvp↔village, economyUi↔ageScreen); add a cycle check to the boundaries test (Q22/Q23). **CODE**
- [ ] applyDefenseLosses swallows sim exceptions and takes all loot (Q33). **CODE**
- [ ] Battle aura and support lookups still per pair per tick (Q28). **CODE**
- [ ] Remove ~20 unreferenced exports, 45 over-exports, unused port interfaces (Q14/Q15). **CODE**
- [ ] Mark generated JSON/MD files as generated; add generator no-op tests (Q5). **CODE**
- [ ] Move slow model-player tests out of the default suite; replace wall-clock ms assertions (Q24/Q25). **CODE**
- [ ] Run PgRepo tests in default CI (Postgres only in the Server release workflow) (Q26). **CODE**
- [ ] Rules-archive digest updated by hand; retire frozen rule snapshots v3–v6 when usage drops (Q4). **CODE**
- [ ] Purge dump.rdb from git history (Q2); clean "Elder Toadstool" / acorn vocabulary in comments and tools; refresh README (Q12/Q13). **CODE**
- [ ] Magic numbers in runCommand / migrate / basegen (Q20). **CODE**

Stale documentation

- [ ] PVP_ROADMAP step 33 says 11 products (14 exist); known gaps 5 and 7 not struck through — verify and update. **DOCS**
- [ ] MARKETS.md says the in-game withdrawal acknowledgement is missing, but economyUi.ts:171–177 now shows it. **DOCS**
- [ ] Record the decision to pursue mainland China through a licensed publisher: docs/legal/MARKETS.md (China row, line 47), docs/localization/CHINESE.md step 3, docs/store/LISTING.md:84. **DOCS**
- [ ] Any closed WORLDWIDE_AUDIT finding (e.g. GLOBAL-016 after the first Postgres CI run) needs its closure record updated. **DOCS**

15. Growth, retention and live-ops features

- [ ] Live-ops: only four distinct events on a 14-day loop; schedule not server-delivered; no events calendar view (G10). **CODE DESIGN**
- [ ] Seasonal and holiday theming (G11). **ART DESIGN**
- [ ] Referral reward and village-image share (G13). **CODE DESIGN**
- [ ] Shareable replays (link or clip); replays are local-only (G14). **CODE**
- [ ] Leaderboards out of Settings; friends/alliance/country tabs; player profile (G15). **CODE**
- [ ] Alliance perks; donation push defaults off (G16). **DESIGN CODE**
- [ ] News / inbox panel (G17). **CODE**
- [ ] Analytics: adults-only opt-in sample, UTC days, catalog-USD revenue, JSON-only staff view; incremental rollup past ~10M events (Growth 1). **CODE**
- [ ] Experiments limited to offer timing; no price/reward tests or significance tooling (Growth 3). **CODE DESIGN**
- [ ] Landing page, Open Graph tags (0 og: tags in index.html), manifest screenshots and shortcuts (G23). **CODE OWNER**

16. Marketing and launch plan

- [ ] Decide the web build's role: browser demo capped at Keep 4 handing off to the app, or full web with web payments (G28). **OWNER**
- [ ] Soft-launch plan: wave A Philippines, Canada, Australia, NZ (8–12 weeks); wave B Brazil, Türkiye, Indonesia after legal; Japan after seller facts. **OWNER**
- [ ] KPIs and kill criteria before spending: tutorial completion, first raid, D1/D7/D30, payer conversion by D7, ARPDAU, crash-free sessions (G30). **OWNER**
- [ ] Track tutorial drop-off at finishKeep once data exists (G4). **OWNER**
- [ ] Budget ~$15–40k paid acquisition plus $5–10k creative (vendor benchmarks — check against Adjust/AppsFlyer). **OWNER**
- [ ] Play pre-registration with a non-random reward and Apple pre-order, 3–6 weeks before launch. **OWNER**
- [ ] Discord, subreddit, short-form video channel — staffed and moderated. **OWNER**
- [ ] Yearly: rotate keys, renew Apple membership, track Billing and Play target-API deadlines. **OWNER**

17. Release gates (from RELEASE.md and TESTING.md — none checked yet)

- [ ] Production server deployed from the same commit, with a passing backup drill. **OWNER**
- [ ] Store products approved; notification URLs registered; listing, privacy label and age rating complete. **OWNER**
- [ ] Release tag and a green "Test, build and deploy" run. **OWNER**
- [ ] Android aab job actually ran (Billing ≥ 8, signature, versionCode, 16 KB). **OWNER DEVICE**
- [ ] iOS SDK accepted, capabilities enabled. **OWNER**

- [ ] VITE_PUBLIC_URL, VITE_GOOGLE_CLIENT_ID, GOOGLE_CLIENT_IDS set. — OWNER
- [ ] TestFlight internal + external, Play closed track 12+ testers × 14 days. — OWNER DEVICE
- [ ] All native device checks in section 6 passed. — DEVICE
- [ ] Contract tests have run against a deployed staging host (never yet). — OWNER

18. Mainland China release (separate track)

The owner wants a mainland China release. It needs a Chinese publishing partner and a China build of the game; expect 6–18 months. Rules below are general knowledge as of 2026 and must be confirmed with the publisher or a China-licensed lawyer.

Publisher and licence (the main obstacle)

- [ ] Find and contract a licensed Chinese game publisher; foreign companies cannot hold the licence or operate the game themselves. Agree revenue share and who runs servers. — OWNER LEGAL
- [ ] Apply (through the publisher) for the NPPA imported-game publishing licence (版号); approvals come in limited monthly batches. — OWNER LEGAL
- [ ] Software copyright registration (软著) for the game. — OWNER LEGAL
- [ ] ICP filing for the app and the server domain (Apple's China storefront requires the licence number and the app filing number). — OWNER LEGAL
- [ ] Prepare a playable China build in Simplified Chinese for the content review. — CODE

Content review

- [ ] Review art and text for review-sensitive content: Japanese military imagery, blood, skeletons, gambling-like features. — DESIGN ART
- [ ] Translate everything still in Latin letters for China: computer players' Japanese names, language names in Settings, US$ prices. — CODE
- [ ] Launch screen shows the licence (ISBN and approval) numbers, the healthy-gaming notice (健康游戏忠告) and the age-suitability label (适龄提示). — CODE
- [ ] Native Chinese review of all game text, tutorial, purchase, safety and legal pages (none done yet). — OWNER

Minors and real-name rules

- [ ] Real-name verification for every player through the national anti-addiction system (publisher connects); replaces the self-declared age screen in the China build. — CODE OWNER
- [ ] Server-enforced play-time limits for under-18s (2021 rules: 8–9 pm on Fridays, weekends and public holidays only) with lock-out and warnings. — CODE
- [ ] China spending caps by age (under 8: none; 8–16: ¥50 per purchase / ¥200 a month; 16–18: ¥100 / ¥400) in RMB, replacing the USD under-18 cap. — CODE
- [ ] Chat and Alliance UGC: real-name players only, Chinese sensitive-word filter (today 30 English words), fast takedown process. — CODE OWNER

Data and servers

- [ ] Separate server and database hosted in mainland China (PIPL data localisation), with an ICP licence for the host. — OWNER CODE
- [ ] China as its own region: separate PvP ladder, Alliances, Gangs and leaderboards, isolated from the worldwide server. — CODE
- [ ] Replace or disable Sentry crash reports (cross-border transfer); local analytics only. — CODE
- [ ] Do not rely on the GitHub Pages web build in China (blocked or unreliable; web games also need the licence). — OWNER
- [ ] PIPL-compliant Chinese privacy policy and terms; cross-border transfer assessment if any data leaves China. — LEGAL

Android without Google

☐	Replace Google Play Billing with the payment SDKs of the Chinese Android stores (Huawei, Xiaomi, OPPO, vivo, Honor, Tencent MyApp) or the publisher's aggregator; Alipay and WeChat Pay.	CODE OWNER
☐	Server receipt checks for each Chinese store's payments.	CODE
☐	Replace Sign in with Google with store accounts or WeChat/QQ login (via the publisher).	CODE
☐	Replace FCM push with vendor push (Huawei/Xiaomi/OPPO/vivo) or an aggregator such as JPush.	CODE
☐	Remove Play Age Signals from the China build (real-name system replaces it).	CODE
☐	Submit to each Chinese Android store (each has its own review, signing and revenue share, often ~50%).	OWNER

Prices, support and operations

☐	RMB price points for iOS China and the Android stores.	OWNER
☐	Chinese-language support and moderation staff with response targets.	OWNER
☐	Keep the China build a variant of the same code (same game rules and battle sim; swapped sign-in, payments, push, age checks and server region) so balance and saves stay in step.	CODE